Zizhuo Xu

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EDUCATION

Huazhong University of Science and Technology *Wuhan, China*
Bachelor of Engineering in Marine Engineering **GPA: 3.39/5.0** *Sep. 2022 – Jun. 2026*
School of Naval Architecture and Ocean Engineering **Average: 79.0/100** Credits: 154.0
Undergraduate Thesis: *Physics-Structured Ship Maneuvering Prediction under Multi-Source Disturbances with Uncertainty Decomposition*
Advisor: Xianzhou Wang

PUBLICATIONS

[1] **Xu, Zizhuo**; Ma, Yinghao; Meng, Junjie; Wang, Xianzhou. **Regime-Gated Identifiability-Guided Expert Specialization of the MMG Model for Generalizable Ship Maneuvering Prediction**. SSRN. https://ssrn.com/abstract=7239757. DOI: 10.2139/ssrn.7239757. Under review at *Ocean Engineering* **(JCR Q1; IF: 6.3).**

[2] **Xu, Zizhuo**; Ma, Yinghao; Wang, Xianzhou. **Efficient Ship Maneuvering Model Identification via Differentiable MMG: Is Full-Parameter Calibration Always Better?** SSRN. https://ssrn.com/abstract=6965894. DOI: 10.2139/ssrn.6965894. Under review at *Ocean Engineering* **(JCR Q1; IF: 6.3).**

[3] **Xu, Z.**; Yao, Z.; Luo, B.; Wang, X. **Physics-Structured Residual Learning for Ship Maneuvering Prediction: Multi-Source Disturbance Decomposition and Compensation**. *Journal of Marine Science and Engineering* 2026, 14, 808. https://doi.org/10.3390/jmse14090808 **(JCR Q2; IF: 3.2).**

RESEARCH EXPERIENCE

Huazhong University of Science and Technology *Jan. 2026 – Present*
Undergraduate research and thesis work supervised by Xianzhou Wang, focused on system identification, data-driven neural-network uncertainty quantification, and differentiable physics-based modeling.

Regime-Gated Identifiability-Guided Expert Specialization of the MMG Model for Generalizable Ship Maneuvering Prediction *Jul. 2026 – Present*

Reframed cross-regime failure as **model-class mismatch:** a single constant-coefficient MMG model cannot represent hydrodynamics that vary with operating state, even when locally well identified.

Characterized a connected **low-dimensional operating-condition manifold** with diffuse boundaries, establishing K as a resolution parameter rather than a fixed physical constant and showing that unrestricted local calibration collapses through drift along weakly identifiable directions.

Developed an identifiability-constrained regime-expert formulation that freezes 21 weakly identifiable parameters and specializes 24 locally supported coefficients, reducing mean 80-s trajectory RMSE from 0.740 to 0.486 m across eight held-out berthing trajectories.

Differentiable MMG for Ship Maneuvering Model Identification *May 2026 – Present*

Challenged the assumption that full-parameter calibration necessarily improves physics-based ship maneuvering models; developed an end-to-end differentiable 45-parameter MMG implementation that reduced calibration time from approximately five days to 20 minutes, enabling systematic parameter-subset studies.

Revealed that maneuvering trajectories constrain only a low-dimensional parameter subspace: the sharply concentrated spectrum of the trajectory Fisher information matrix exposed severe overparameterization and weakly identifiable directions.

Exposed a reversal between fitting and generalization: a Fisher-selected 13-parameter model reduced berthing RMSE from 1.44 to 1.19 m despite a higher identification loss, showing that parameter-subset selection must balance information coverage against overfitting along weakly constrained directions.

Physics-Structured Ship Maneuvering Prediction under Multi-Source Disturbances with Hierarchical UQ *Jan. 2026 – Apr. 2026*

Moved beyond monolithic residual correction by separating physically heterogeneous disturbances: built an MMG-anchored modular framework in which a CNN-SE-BiLSTM encoder infers slowly varying shallow-water, hull-fouling, and rudder-degradation effects, while measured wind enters through a dedicated pathway.

Extended the framework from prediction to hierarchical uncertainty attribution: used MC Dropout and the law of total variance to decompose predictive uncertainty across the encoder, disturbance-specific experts, upstream propagation, and individual motion degrees of freedom.

Validated the framework under strict long-horizon closed-loop rollout: reduced position RMSE by 74.7%–91.7%, including from 255.3 to 8.3 m under severe shallow-water conditions and from 227.7 to 14.4 m under triple-source disturbances.

RESEARCH INTERESTS

Modeling and Control of Dynamical Systems

System identification, state and parameter estimation, real-time model predictive control, and physics-based, data-driven, and hybrid modeling for vehicle, marine, and energy systems.

Neural Networks for Sensor-Based Monitoring and Fault Diagnosis

Representation learning from acoustic, vibration, and other time-series sensor data for condition monitoring, anomaly detection, and fault diagnosis.

ENGINEERING PROJECTS

Intelligent Waste-Sorting Bin — Project Lead *Sep. 2024 – Dec. 2024*

Led mechanical development, including SolidWorks design of the bin enclosure, robotic arm, and sliding platform, as well as component procurement and assembly.

Contributed to dataset preparation and training of a YOLOv8-based visual recognition model.

Built and tested the prototype, successfully achieving the intended waste classification and sorting functions.

Engineering Drawing and Product Modeling Competition *Mar. 2024 – Jul. 2024*

Created 3D part models from engineering drawings and assembled them into a complete mechanical assembly using SolidWorks.

Designed a functional 3D mechanical model that met the specified structural and operational requirements.

Completed the modeling, assembly, and engineering drawing tasks within the competition time limit.

SKILLS

Neural Networks and Uncertainty Quantification: PyTorch; MLPs, CNNs, LSTMs, and hybrid neural architectures; MC Dropout, deep ensembles.

System Identification and State Estimation: Least-squares and ridge regression, extended and unscented Kalman filtering (EKF/UKF), and differentiable parameter identification for nonlinear dynamical systems.

CAD and Prototyping: SolidWorks and AutoCAD, 3D part and assembly modeling, engineering drawings, component assembly, and prototype testing.

Scientific Computing and Research Tools: Python, PyTorch, NumPy, SciPy, and Git.

Languages: Chinese (Native); English (IELTS 7.0).

AWARDS

17th "Higher Education Cup" National Advanced Graphics Competition — **First Prize, Mechanical Advanced Graphics Track (Individual; ranked 23rd nationally)** *Jul. 2024*

17th "Higher Education Cup" National Advanced Graphics Competition — **First Prize, Mechanical Team (team ranked 1st nationally)** *Jul. 2024*

4th Hubei Provincial Advanced Graphics and Product Information Modeling Competition — **First Prize, Mechanical Category (ranked 7th in Hubei Province)** *Jun. 2024*